

5.3 Independence of Path; Conservative Vector Fields

We begin this section with an example which will set the stage for what we will be discussing.

Example 1: Suppose C_1 is given by $\mathbf{r}_1(t) = \langle t, t \rangle$ where $0 \leq t \leq 1$. Suppose C_2 is given by $\mathbf{r}_2(t) = \langle t, t^2 \rangle$ where $0 \leq t \leq 1$.

a) Where do the above curves start and end?

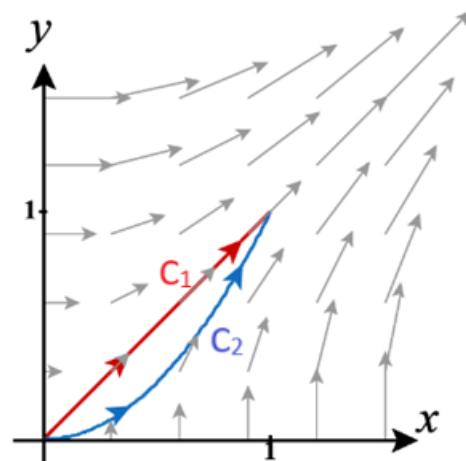
By plugging in 0 for t and 1 for t it is easy to see that both curves start at $(0,0)$ and both curves end at $(1,1)$.

b) Show $\int_{C_1} ydx + xdy = \int_{C_2} ydx + xdy$.

In the case of C_1 we have $x = t$ $y = t$ which implies that $dx = dt$ and $dy = dt$. This means $\int_{C_1} ydx + xdy = \int_0^1 tdt + \int_0^1 tdt = \int_0^1 2tdt = t^2 \Big|_0^1 = 1$. In the

case of C_2 we have $x = t$ $y = t^2$ which implies that $dx = dt$ and $dy = 2tdt$. This means $\int_{C_2} ydx + xdy = \int_0^1 t^2 dt + \int_0^1 2t^2 dt = \int_0^1 3t^2 dt = t^3 \Big|_0^1 = 1$.

Thus, $\int_{C_1} ydx + xdy = \int_{C_2} ydx + xdy$.



Question: If we were to use any smooth curve, $\mathbf{r}(t)$ from $(0,0)$ to $(1,1)$, would we still get that $\int_C \langle y, x \rangle \cdot d\mathbf{r} = 1$?

Questions like the one posed above are the theme of this section. Surprisingly it is the case that for some vector fields the path that we take from point A to point B has *no effect* on the value of the resulting line integral! This is a FASCINATING idea especially considering the fact that there are infinite number of paths from A to B and the complexity and length of some of these paths can be astounding!

So, if we have the right kind of vector field computing a line integral will ultimately only depend on the starting point and ending point for our object. What do we mean though by the “right kind of vector field”. It turns out that in order for the path of motion to not have any affect on the resulting line integral we simply must be working with a conservative vector field. The proof of the next theorem shows us why this is the case.

Theorem (Fundamental Theorem of Line Integrals):

Let C be a smooth curve given by a vector function $\mathbf{r}(t)$ where $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector $\bar{\nabla} f$ is continuous on C . Then,

$$\int_C \bar{\nabla} f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

Note: This result is amazing! It says that if our vector field of interest is conservative (i.e. is the gradient of some scalar function) all we need to know about the curve to compute the line integral is the endpoints!

Proof (may be omitted from class): We will prove this in the case that $\mathbf{r}$ has three components. We know that

by definition:
$$\int_C \bar{\nabla} f \cdot d\mathbf{r} = \int_a^b \bar{\nabla} f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_a^b \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle dt = \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt.$$

By the chain rule we know that $\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = \frac{d}{dt} f(\mathbf{r}(t))$. So the later integral becomes

$$\int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt.$$
 Now, we can think of $f(\mathbf{r}(t))$ as function which maps a real number t to some real number.

So, by the fundamental theorem of calculus
$$\int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

■

Fact: One can generalize the above argument and show that the above result also holds if C is a piecewise smooth curve.

Example 2: Find the work done by $\mathbf{F}(x, y) = \langle y, x \rangle$ on a particle which moves along some piecewise smooth curve C that starts at $(0, 0)$ and ends at $(1, 1)$.

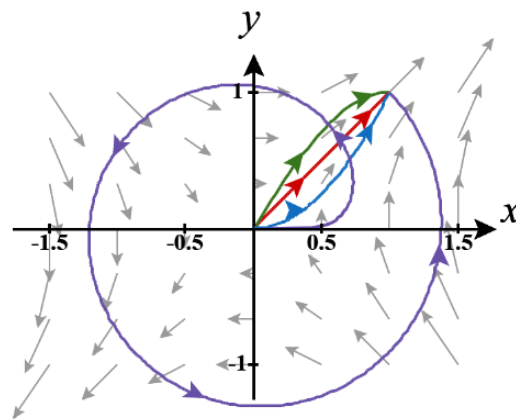
We know that $W = \int_C \mathbf{F} \cdot d\mathbf{r}$. It is impossible to find this line integral

with the ideas of section 5.2. This is because it is impossible to find a parameterization of C since we are only given the starting and ending points for C .

Instead we notice that if $f(x, y) = xy$, then

$\bar{\nabla} f(x, y) = \langle y, x \rangle = \mathbf{F}(x, y)$. So, by the fundamental theorem of line integrals we have that $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \bar{\nabla} f \cdot d\mathbf{r} = f(1, 1) - f(0, 0) = 1$.

Note that the paths in the picture above could all possibly represent the curve C and are also not the only paths that could be the same as C .



Note: When applying the fundamental theorem of line integrals the most challenging part is finding a scalar function whose gradient equals the vector field of interest (in some cases such a scalar function might not even exist). This challenge will be discussed in much more detail later in this section.

Making Connections

Recall that a line integral was interested in “adding up” a quantity as we moved along a curve. It is easy to see that this concept extends upon the idea of definite integrals from Calculus I.

Definite Integral vs Line Integrals:

In Calculus I we had a function $f(x)$ and we were interested in “adding up” the values of this function over an interval $[a, b]$. So the integral $\int_a^b f(x) dx$ is very similar to the line integral $\int_C f(x, y) ds$.

- The functions $f(x)$ and $f(x, y)$ represent the quantities that we are adding
- The interval $[a, b]$ and the curve C represent the path that we are moving along.

Fundamental Theorem of Calculus vs FTOLI:

Remember that in Calculus I the Fundamental theorem showed us that to find $\int_a^b f(x) dx$ we can simply look at the behavior of an antiderivative at the end points of the interval $[a, b]$. Note that the FTOLI states that we can evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ by evaluating the “anti-gradient” of $\mathbf{F}$ at the endpoints of C !

- $\int_a^b f(x) dx = F(b) - F(a)$
- $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

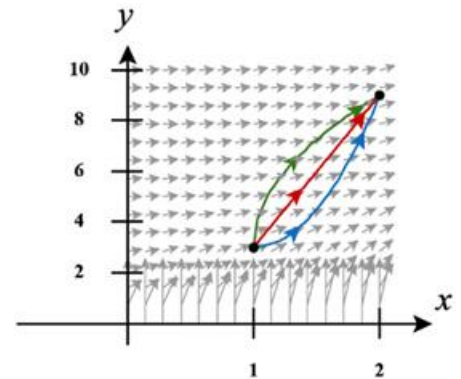
It is AMAZING that this concept of finding an integral by evaluating a specific function at some endpoints extends across definite integrals and line integrals, however we will see even later in Chapter 5 how this concept continues to extend to specific double and triple integrals!

Example 3: Suppose C is a smooth curve which starts at $(1, 3)$ and ends at $(2, 9)$. Suppose

$\mathbf{F}(x, y) = \langle y^2 e^{xy}, e^{xy}(1 + xy) \rangle$. Find the exact value of $\int_C \mathbf{F} \cdot d\mathbf{r}$.

We note that if $f(x, y) = ye^{xy}$, then $\nabla f(x, y) = \mathbf{F}(x, y)$. So, by the fundamental theorem of line integrals

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(2, 9) - f(1, 3) = 9e^{18} - 3e^3.$$



We now discuss the notion of independence of path which will be of great importance for the rest of the section.

Definition: We say that C is a path if it is a piecewise-smooth curve.

Definition: Suppose that $\mathbf{F}$ is a continuous vector field with domain D . We say that the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path if $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ for any two paths C_1 and C_2 in D that have the same initial and terminal points.

Example 4: Suppose that $f(x, y, z)$ is a scalar function such that $\mathbf{F} = \vec{\nabla} f$ is continuous on the domain D .

TRUE or FALSE: $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path.

This is TRUE. To see why, take any two paths C_1 and C_2 in D which start and end at the same point.

Specifically, suppose that both paths start at (a, b, c) and both paths end at (k, m, n) . Then, by the fundamental theorem of line integrals:

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \vec{\nabla} f \cdot d\mathbf{r} = f(k, m, n) - f(a, b, c)$$

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \vec{\nabla} f \cdot d\mathbf{r} = f(k, m, n) - f(a, b, c).$$

Hence, $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ and $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path by definition.

*Note: Another way to say the above result is that **line integrals of conservative vector fields are independent of path**. A really good question to ask from here is: are all vector fields which are independent of path conservative?*

So we have established the following fact.

Theorem (5.3.1): Suppose $\mathbf{F}$ is a continuous vector field with domain D . If $\mathbf{F}$ is a conservative vector field, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D .

Note: The contrapositive of this statement is logically equivalent to the statement itself and hence is also true: Suppose $\mathbf{F}$ is a continuous vector field with domain D . If $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not independent of path in D , then $\mathbf{F}$ is not a conservative vector field.

Example 5: Suppose C_1 is given by $\mathbf{r}_1(t) = \langle t, t \rangle$ where $0 \leq t \leq 1$. Suppose C_2 is given by $\mathbf{r}_2(t) = \langle t, t^2 \rangle$ where $0 \leq t \leq 1$. Suppose that $\mathbf{F}(x, y) = \langle y^2, x^2 \rangle$.

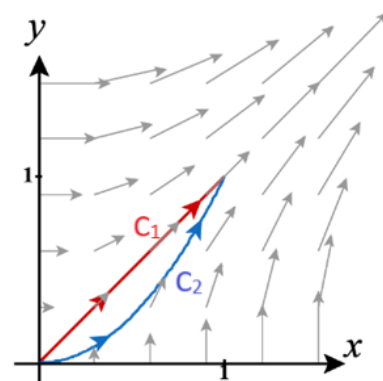
a) Find $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$.

We note that

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 \langle t^2, t^2 \rangle \cdot \langle 1, 1 \rangle dt = \int_0^1 2t^2 dt = \left. \frac{2}{3} t^3 \right|_0^1 = \frac{2}{3}.$$

b) Find $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$.

We note that $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 \langle t^4, t^2 \rangle \cdot \langle 1, 2t \rangle dt = \int_0^1 (t^4 + 2t^3) dt = \left. \left(\frac{t^5}{5} + \frac{t^4}{2} \right) \right|_0^1 = \frac{1}{5} + \frac{1}{2} = \frac{7}{10}.$



c) Is $\int_C \mathbf{F} \cdot d\mathbf{r}$ independent of path in 2 space? Explain.

We note that the curves C_1 and C_2 start at the same point and end at the same point in 2 space. Since we got different answers in parts (a) and (b) we know that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not independent of path in 2 space by definition.

d) Is the above vector field conservative? Explain.

We know from the above fact that if $\mathbf{F}$ is conservative then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in the domain of $\mathbf{F}$.

Since we know that in this case $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not independent of path, it follows that $\mathbf{F}$ is not conservative (Note:

We are using the contrapositive here).

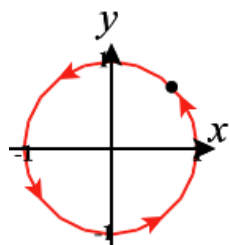
Note: The idea in the above example is crucial. Specifically, it shows how we can prove a vector field is not conservative. This means in the above example it is impossible to find a scalar function whose gradient is equal to $\mathbf{F}$.

We have now seen that *conservative vector fields* have some very useful properties and can be used to make the calculation of line integrals much easier.

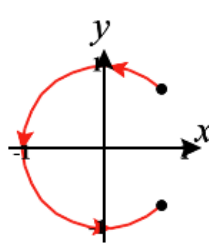
So, being able to determine if a vector field is conservative is now of great importance to us and we will spend the rest of this section trying to pursue this.

Definition: A curve is called closed if its terminal point coincides with its initial point.

Closed



Non-closed



Theorem (5.3.2): Suppose that $\mathbf{F}$ is a continuous vector field with domain D . $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path in D .

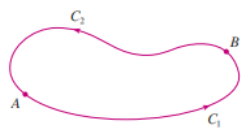
Proof: (Shown on next page. Not done in class) See the argument posted in the Blackboard notes. I highly recommend reading this argument until you completely understand it.

We start by showing that if we know that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D then we must have $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path in D .

Suppose that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D and C is any closed path in D . We can choose any two points



A and B on C and regard C as being composed of the path C_1 (from A to B) followed by the path C_2 (from B to A). See the picture below.



We can then see that $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{-C_2} \mathbf{F} \cdot d\mathbf{r} = 0$ since C_1 and $-C_2$ have the same initial and terminal points.

Now we must show that if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path in D then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D .

If we know that $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ whenever C is a closed path in D , then notice that if we take any two paths, say C_1 and C_2 , from A to B in D , and define C to be the curve consisting of C_1 followed by $-C_2$ we obtain the following:

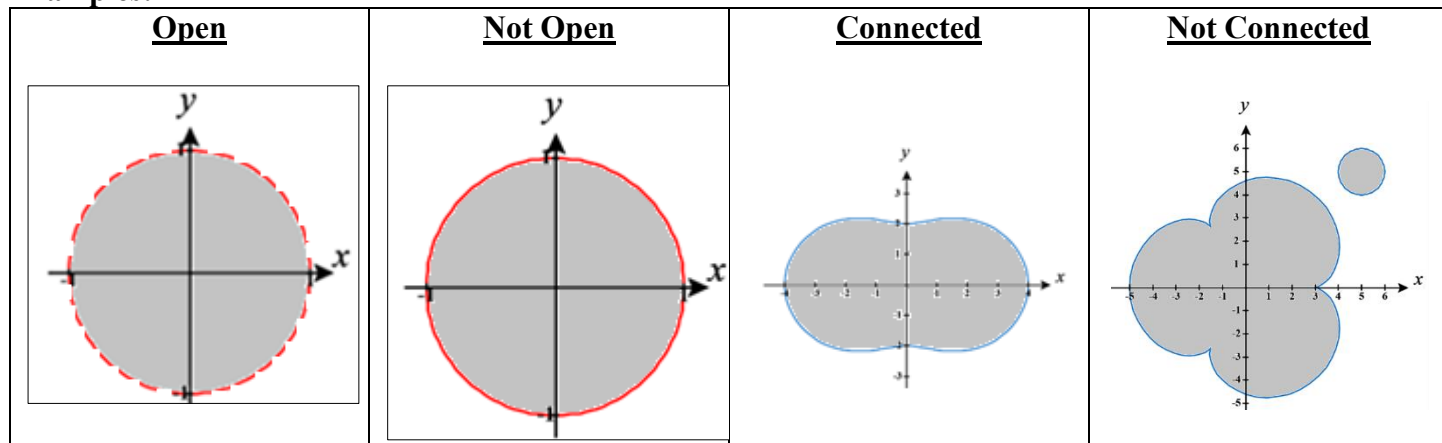
$$0 = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{-C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$$

This, of course, implies that $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ and therefore we have independence of path.

Definition: We say that a region D is open if for every point P in D there is a disk with center P that lies entirely in D . Intuitively, D is open if it does not contain any of its boundary points.

Definition: We say that a region D is connected if any two points in D can be joined by a path that lies in D .

Examples:



Theorem (5.3.3): Suppose $\mathbf{F}$ is a vector field that is continuous on an open connected region D . If $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D , then $\mathbf{F}$ is a conservative vector field on D ; that is, there exists a function f such that $\nabla f = \mathbf{F}$ (we say that f is a potential function for $\mathbf{F}$).

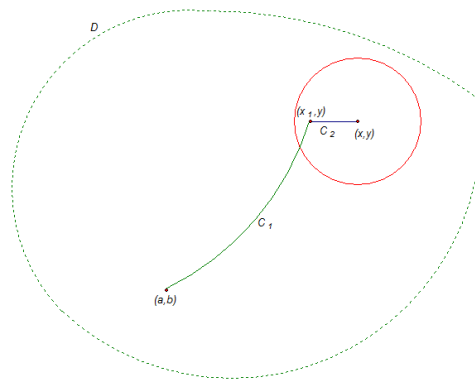
Proof: We will prove this in the case that $\mathbf{F}$ has two components (the case where $\mathbf{F}$ has three components is similar). We need to construct a potential function for $\mathbf{F}$. Suppose (a,b) is a fixed point in D . For any point

$$(x,y) \text{ in } D \text{ we let } f(x,y) = \int_{(a,b)}^{(x,y)} \mathbf{F} \cdot d\mathbf{r}.$$

The precise meaning of the integral $\int_{(a,b)}^{(x,y)} \mathbf{F} \cdot d\mathbf{r}$ is that it represents the line integral of $\mathbf{F}$ along any path C which is contained in D and which starts at (a,b) and ends at (x,y) . Since $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D , there

is one unique value associated with $\int_{(a,b)}^{(x,y)} \mathbf{F} \cdot d\mathbf{r}$.

Now, since D is open we know that there exists a disk contained in D with center (x,y) . Choose any point (x_1,y) in the disk with $x_1 < x$ and let C consist of any path C_1 from (a,b) to (x_1,y) followed by the horizontal line segment from (x_1,y) to (x,y) (these paths must exist since D is connected). A possible picture is below.



Now, we notice that $f(x, y) = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{(a,b)}^{(x_1,y)} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$.

Since the first integral does not depend on x we have that $\frac{\partial}{\partial x} f(x, y) = 0 + \frac{\partial}{\partial x} \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$.

Now, a parameterization for C_2 is $\mathbf{r}(t) = \langle t, y \rangle$ where $x_1 \leq t \leq x$. Since we supposed that $\mathbf{F}$ has two components we know that $\mathbf{F}$ is of the form $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$. So, we have that

$$\frac{\partial}{\partial x} \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \frac{\partial}{\partial x} \int_{x_1}^x \langle P(t, y), Q(t, y) \rangle \cdot \langle 1, 0 \rangle dt = \frac{\partial}{\partial x} \int_{x_1}^x P(t, y) dt = P(x, y). \text{ Thus, } \frac{\partial}{\partial x} f(x, y) = P(x, y).$$

Similarly, one can show that $\frac{\partial}{\partial y} f(x, y) = Q(x, y)$. This means that $\vec{\nabla} f = \mathbf{F}$ and our proof is complete. ■

Theorem (5.3.4): If $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$ is a conservative vector field, where P and Q have continuous first order partial derivatives on a domain D , then throughout D we have $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Question: Why does this theorem hold? (Hint: Think about the theorem from chapter 3 which told us when $f_{xy}(a, b) = f_{yx}(a, b)$)

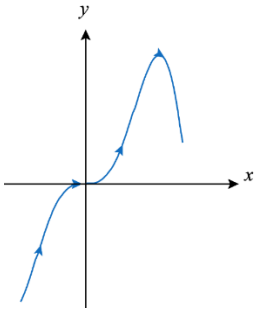
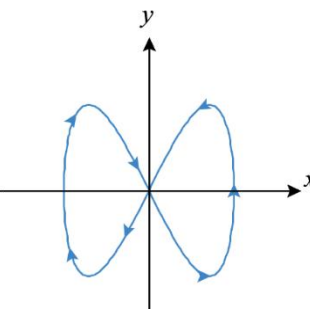
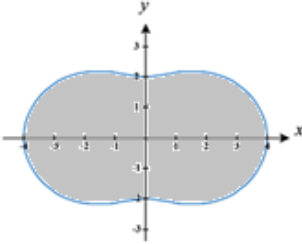
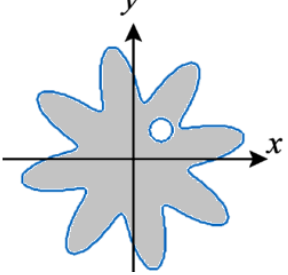
Note: From here the crucial question is: Does the converse of the above theorem hold? If it does, we would have a really nice way to check if a two dimensional vector field is conservative.

Definition: A simple curve is a curve which doesn't intersect itself anywhere between its endpoints.

Definition: A simply-connected region in the plane is a connected region D such that every simple closed curve in D encloses only points in D .

Note: Intuitively, a simply-connected region is a connected region which contains no holes.

Examples:

<u>Simple Curve</u>	<u>Non-Simple Curve</u>	<u>Simply Connected Region</u>	<u>Not Simply Connected Region</u>
			

Theorem (5.3.5): Let $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$ be a vector field on an open simply-connected region D . Suppose that P and Q have continuous first order partial derivatives and $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$. Then, $\mathbf{F}$ is conservative.

Note: Since 2-space is an open simply-connected region, the above two theorems provide us with a very useful way for checking whether or not a vector field is conservative.

Note: An easy way to check if a vector field with three components is conservative that is similar to the above results will be shown later.

Example 6: Determine whether or not $\mathbf{F}(x, y) = \langle x - y, x - 2 \rangle$ conservative.

Since $\mathbf{F}$ is defined on $\mathbb{R}^2$ and the components of $\mathbf{F}$ have continuous first order partial derivatives, the two most recent theorems apply. We note that $\frac{\partial P}{\partial y} = -1$ and $\frac{\partial Q}{\partial x} = 1$. Since $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ we know that $\mathbf{F}$ is not conservative.

Example 7:

a) Show that $\mathbf{F}(x, y) = (3x^2 - 2y^2)\mathbf{i} + (-4xy + 3)\mathbf{j}$ is conservative.

We note that $\mathbf{F}$ is defined on $\mathbb{R}^2$ and the components of $\mathbf{F}$ have continuous first order partial derivatives. So, the two most recent theorems apply. Then, $\frac{\partial P}{\partial y} = -4y = \frac{\partial Q}{\partial x}$, and we know by the most recent theorem that $\mathbf{F}$ is conservative.

b) Find a potential function for $\mathbf{F}$.

We seek a function f such that $\vec{\nabla} f = \mathbf{F}$. So, we know that f satisfies $f_x(x, y) = 3x^2 - 2y^2$ and $f_y(x, y) = -4xy + 3$.

If we integrate both sides of the first equation with respect to x , we get

$$\int f_x(x, y) dx = \int (3x^2 - 2y^2) dx \Leftrightarrow f(x, y) = x^3 - 2y^2x + h(y).$$

Then, if we differentiate both sides of the latter equation with respect to y we get $f_y(x, y) = -4yx + h'(y)$, but we already know that $f_y(x, y) = -4xy + 3$.

This means that $h'(y) = 3$, and if we integrate both sides of this equation with respect to y , we get $h(y) = 3y + C$ where C is some constant.

If we let this constant be 0, we see that a potential function for $\mathbf{F}$ is $f(x, y) = x^3 - 2y^2x + 3y$.

- c) Suppose that the curve C is given by $\mathbf{r}(t) = \langle e^t \sin(t), e^t \cos(t) \rangle$ where $0 \leq t \leq \pi$. Find the work done by $\mathbf{F}$ in moving a particle along C .

Luckily since we have a potential function for $\mathbf{F}$ we can do this question with the fundamental theorem of line integrals rather than the tedious methods of section 5.2!

We note that C starts at the point $(0,1)$ (to see these find $\mathbf{r}(0)$) and it ends at the point $(0, -e^\pi)$ (to see this find $\mathbf{r}(\pi)$).

So, by the fundamental theorem of line integrals $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(0, -e^\pi) - f(0, 1) = -3e^\pi - 3$.

Note: The method of presented in the solution to part (b) of the above example can be generalized to vector fields with three components when we know that we have a conservative vector field. In the next example assume that the vector field is conservative and try to generalize the methods of part (b).

Example 8 (if time permits): Find a potential function for $\mathbf{F}(x, y, z) = \langle yz + z, xz + 1, xy + x \rangle$.

We seek a three variable function f so that $f_x(x, y, z) = yz + z$ $f_y(x, y, z) = xz + 1$ $f_z(x, y, z) = xy + x$.

If we integrate both sides of the first of these equations with respect to x we get

$$\int f_x(x, y, z) dx = \int (yz + z) dx \Leftrightarrow f(x, y, z) = xyz + zx + g(y, z).$$

If we differentiate both sides of this equation with respect to y we get $f_y(x, y, z) = xz + g_y(y, z)$, but we already know that $f_y(x, y, z) = xz + 1$. So, we have that $g_y(y, z) = 1$.

If we integrate both sides of this equation with respect to y we get $\int g_y(y, z) dy = \int 1 dy \Leftrightarrow g(y, z) = y + h(z)$.

If we plug this result into $f(x, y, z) = xyz + zx + g(y, z)$, we get $f(x, y, z) = xyz + zx + y + h(z)$.

This means that $f_z(x, y, z) = xy + x + h'(z)$, but we already know that $f_z(x, y, z) = xy + x$.

Thus, $h'(z) = 0 \Leftrightarrow \int h'(z) dz = \int 0 dz \Leftrightarrow h(z) = C$ where C is a constant.

If we take this constant to be 0, we see that a potential function for $\mathbf{F}$ is $f(x, y, z) = xyz + zx + y$.